Fiscal Year:	FY 2015	**Task Last Updated:**	FY 07/13/2015
PI Name:	Wyatt, Sarah E Ph.D.		
Project Title:	Proteomics Analysis of Arabidopsis Seedlings in Microgravity		
Division Name:	Space Biology		
Program/Discipline:	SPACE BIOLOGY		
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Developmental biology		
Joint Agency Name:	**TechPort:**		No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Plant Biology		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	45701	**Congressional District:**	15
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	2012 Space Biology NNH12ZTT001N
Start Date:	09/01/2013	**End Date:**	08/31/2016
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	1
No. of Master's Candidates:	3	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA KSC
Contact Monitor:	Levine, Howard	**Contact Phone:**	321-861-3502
Contact Email:	howard.g.levine@nasa.gov		
Flight Program:	ISS		
Flight Assignment:	ISS Rapid Turn Around Flight NOTE: End date is now 8/31/2016 per NSSC information and PI (Ed., 10/8/15)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Luesse, Darron Ph.D. (Southern Illinois University, Edwardsville)		
Grant/Contract No.:	NNX13AM48G		
Performance Goal No.:			
Performance Goal Text:			
Task Description:	The space flight environment provides a unique environment to understand how gravity informs plant growth, development, and physiological processes. The central objective of this study is to determine what proteins are differentially expressed during space flight in developing Arabidopsis seedlings. A body of research has evaluated the differential gene expression on space flight vs ground controls, but gene expression provides an indirect measure of what proteins are expressed. Genes may be transcribed but the transcripts degraded, or translated proteins may be regulated through post translational modifications. Our hypothesis is simple: Some components of plant physiology that are informed by gravity are regulated by post transcriptional or post translational mechanisms. Proteomic analysis is the best/only method to identify these components. We propose to use BRIC-PDFU (Biological Research In Canisters - Petri Dish Fixation Unit) hardware on a Rapid Turn-Around Space Flight Experiment to gain insights into differences in protein profiles between Arabidopsis seedlings grown during space flight and ground based controls. Arabidopsis seedlings will be germinated, grown, and tissues fixed on orbit. Once returned to Earth, total protein will be extracted, labeled with iTRAQ reagents, and analyzed for differences in protein profiles. These data will complement the gene expression data currently available from space flight		

	experiments and also provide novel insights by supplying data on differences in post-transcriptional regulation. They will also provide a foundation for a network analysis to generate additional hypotheses as to the mechanisms involved in gravity perception/response in plants.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This research may lead to the discovery of novel components that determine how plants respond to gravity and to the space flight environment. It will also form a foundation for the establishment of hypotheses to support research on plant improvement for space flight and life support in the space flight environment.
Task Progress:	In the past year, we have completed the Payload Verification Test and actually flew the experiment on SpaceX5, launching Jan 10, 2015. The payload was returned to us Feb 10, 2015. Since that time, we have extracted both the RNA and membrane and soluble proteins from both the space flight samples and ground controls. We also set up additional ground controls to test and control for additional variables. We had good yield on all extractions and the RNA and proteins samples have been sent to the respective Genomics/Proteomics Facilities for processing. We are anxiously awaiting the return of the data, which is expected mid summer.
Bibliography Type:	Description: (Last Updated: 11/25/2025)
Abstracts for Journals and Proceedings	Basu P, Wyatt SE. "Proteomics analysis of plants subjected to modification in the gravity vector." 30th Annual Meeting of the American Society for Gravitational and Space Research, Pasadena, CA, October 22-26, 2014. 30th Annual Meeting of the American Society for Gravitational and Space Research, Pasadena, CA, October 22-26, 2014. , Oct-2014
Abstracts for Journals and Proceedings	Wyatt SE, Shen K, Hayden M. "Plant gravitropic signal transduction: A network analysis leads to gene discovery." 30th Annual Meeting of the American Society for Gravitational and Space Research, Pasadena, CA, October 22-26, 2014. Program for the 30th Annual Meeting of the American Society for Gravitational and Space Research, Pasadena, CA, October 22-26, 2014. , Oct-2014
Abstracts for Journals and Proceedings	Hayden M, Wyatt SE. "Creating a Foundational Gene Network: a Systems Approach for Understanding the Plant Response to Spaceflight." 30th Annual Meeting of the American Society for Gravitational and Space Research, Pasadena, CA, October 22-26, 2014. Program: 30th Annual Meeting of the American Society for Gravitational and Space Research, Pasadena, CA, October 22-26, 2014. , Oct-2014
Abstracts for Journals and Proceedings	Basu P, Luesse DR, Wyatt SE. "Effect of RNAlater® on Proteomic Analysis of Arabidopsis Seedlings." American Society of Plant Biologists (ASPB) Midwestern Section Annual Meeting, St. Louis, MO, March 21-22, 2015. Program. American Society of Plant Biologists (ASPB) Midwestern Section Annual Meeting, St. Louis, MO, March 21-22, 2015. Abstract P16. Search program: http://c.ymcdn.com/sites/my.aspb.org/resource/group/d431703a-d18e-4852-a6a7-fd0622263d6d/Meeting_2015/ASPB_Midwest_2015_Program.pdf ; accessed 7/14/15. , Mar-2015
Abstracts for Journals and Proceedings	Basu P, Hutchinson S, Luesse DR, Wyatt SE. "Preparing for Proteomics Analysis of Arabidopsis Seedlings Grown In Microgravity." Presented at the American Society of Plant Biologists (ASPB) Midwestern Section Annual Meeting, St. Louis, MO, March 21-22, 2015. Program. American Society of Plant Biologists (ASPB) Midwestern Section Annual Meeting, St. Louis, MO, March 21-22, 2015. Abstract T25. Search program: http://c.ymcdn.com/sites/my.aspb.org/resource/group/d431703a-d18e-4852-a6a7-fd0622263d6d/Meeting_2015/ASPB_Midwest_2015_Program.pdf ; accessed 7/14/15. , Mar-2015
Abstracts for Journals and Proceedings	Hutchinson S, Basu P, Wyatt SE, Luesse DR. "Preparation for Proteomic Analysis of Microgravity-Grown Arabidopsis Seedlings" Presented at the American Society of Plant Biologists (ASPB) Midwestern Section Annual Meeting, St. Louis, MO, March 21-22, 2015. Program. American Society of Plant Biologists (ASPB) Midwestern Section Annual Meeting, St. Louis, MO, March 21-22, 2015. Abstract P50. Search program: http://c.ymcdn.com/sites/my.aspb.org/resource/group/d431703a-d18e-4852-a6a7-fd0622263d6d/Meeting_2015/ASPB_Midwest_2015_Program.pdf ; accessed 7/14/15. , Mar-2015
Articles in Peer-reviewed Journals	Basu P, Luesse DR, Wyatt SE. "Proteomic approaches and their application to plant gravitropism." Methods Mol Biol. 2015;1309:119-32. http://dx.doi.org/10.1007/978-1-4939-2697-8_10 ; PubMed PMID: 25981772 , May-2015
Dissertations and Theses	Hutchinson S. "Preparation for Proteomic Analysis of Microgravity-Grown Arabidopsis Seedlings." M.S. Thesis, Southern Illinois University Edwardsville, July 2015. , Jul-2015